Altium

Getting Started

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University of Glasgow RACING

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– FOR INTERNAL USE ONLY –

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1 Introduction

This is the UGR getting started guide for Altium. In December 2025 the decision was made to make the switch from KiCAD (a FOSS hobbyist project) to Altium (industry standard proprietary software). We hope that being able to use this software gives UGR members the preparation they need going into industry having had hands on experience with the exact software used at companies around the world. A sponsorship was obtained giving the team access to the company's online project management software Altium 365 workspace where each design will be stored. This is a move away from the old ways of using the fileshare to host our KiCAD projects, and should streamline our processes.

Altium is actually very well documented in the form of tutorial videos and wikis. This guide will show you where to go to find this information, and lay out some UGR standard practices that may differ from available documentation.

2 Licensing

Whilst the team has a sponsorship for a number of licenses, these are being kept reserved for any members/advisors that do not have access to the standard Altium Student license. Therefore, any information regarding software licensing in this documentation applies to those eligible for the student license (anyone with a student email). If you **do not have access** to Altium Student please make yourself known to your team head, who will prepare the alternative licensing method for you.

3 Setup

3.1 Altium Student Enrollment

For the vast majority of UGR electronics members, we will make use of the standard Altium Student licensing. This license allows anyone enrolled at a valid university course to have access to the Altium Designer Professional software. As such, the enrolling onto this is pretty simple. Use the following link to enroll into their programme, and create an account.

> **Note**
>
> Make sure to use your student email when signing up.

https://www.altium.com/education/students

3.2 Installation

Use the instructions provided in your email from Altium, to install Altium Designer.

The Altium licenses that students have access to are what are called ”On-demand licenses”. These types of licenses are tied to your account and can be transferred between computers but only ever used by one computer at a time.

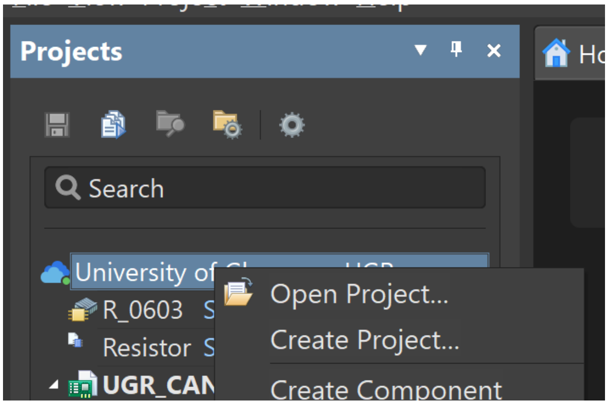
> **Important**
>
> A single license must be **released** before it can be used on another computer. Try to use `Roam` wherever possible, and release the license when you stop using it on that computer.

4 Starting a Design

If the project has already been set up for you, skip to 4.2

4.1 Creating a project

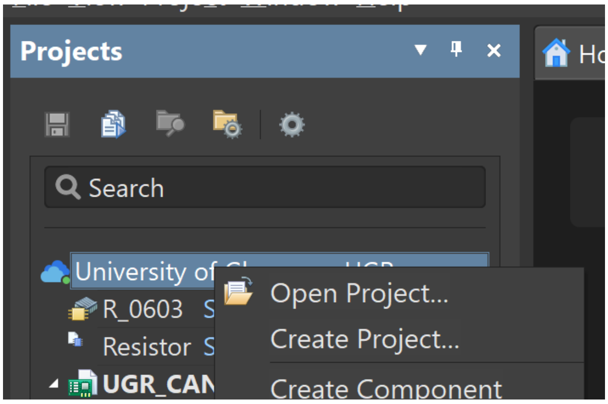
To start, make sure you have the UGRacing workspace open, by checking the top right corner. You will either have a blue cloud with your name/UGRacing, or a grey cloud with the text "Not connected". Select this and choose "University of Glasgow - UGRacing". To create a new project you'll need to use the "Projects" menu on the left hand side. In this panel, right click the blue cloud with the text "University of Glasgow - UGRacing" and choose "Create Project".



In the new window that pops up, fill in the required information, and click create. Ensure the location is still set to "University of Glasgow" on the left hand side, and that version control is checked.

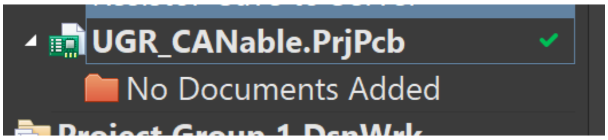
4.2 Opening a project

To open an existing project, use right click the University of Glasgow - UGRacing blue cloud in the Projects panel and click open project.

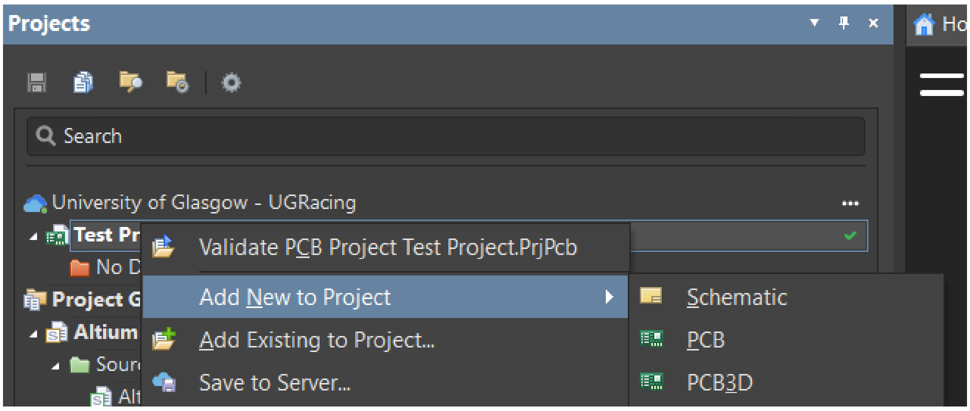


4.3 Starting your design

Once you have opened or created a project you should see an empty project in the "Projects" panel on the left hand side.

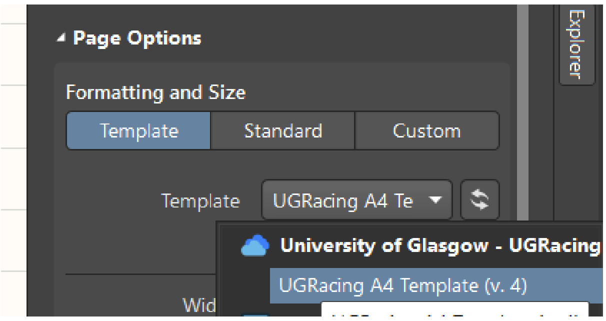


To create a new file right click on your project, and click "Schematic" under "Add new to project".



Now that you have created a schematic, rename it to whatever is appropriate. Next, open the properties panel using the button on the right hand side. If Properties is not there, select it using the "Panels" button in the bottom right hand corner.

Scroll down to the bottom of this panel and under "Page Options" select template, and use the dropdown menu to select "UGRacing A4 Template".



5 Components

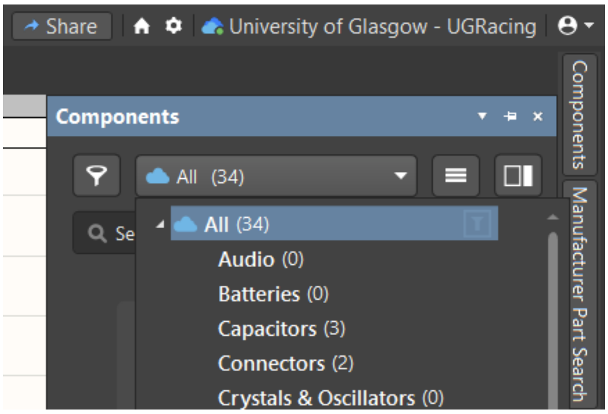
In UGR we have a set of standard components, which are expected to be regularly stocked. Sticking to these components wherever possible means that not only do we save money as a team, but should any repairs or spares need to be assembled we do not need to wait for parts to arrive.

> **Important**
>
> In your designs **do not** use ”Manufacturer Parts Search” results directly, they must be added to the library first. Though feel free to use it to find a suitable part.

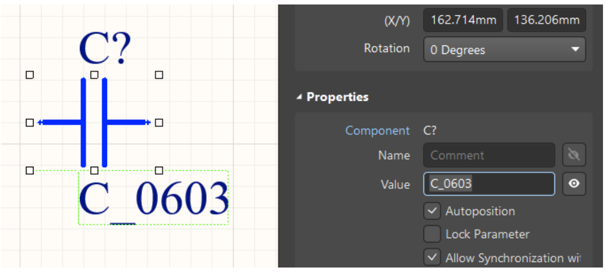
Think of this as a 3 tier system, with Standard Components at the top, Specialised Components in the middle and brand new components at the bottom. If you can’t find what you want in the list of standard components, look in the specialised components, and only if you still can’t find what you want, should you spec and find a new part to be ordered.

5.1 Standard Components

To access these components, open the ”Components” panel on the right hand side, and use the dropdown to find your component.



Once you find your component, you can click and drag it onto your schematic. For passives we have ”Generic” components defined by their package size. To use these, drag them onto the schematic like any other component and double click the package size you chose and edit the ”Value” field to define your value.



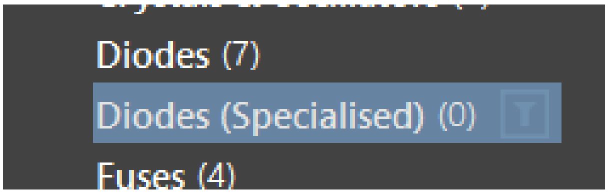
> **Important**
>
> When choosing passive values, refer to the standardisation document. For resistors and capacitors you must only choose values available in the books or on our reels. Any other value should have a specific part specced and found.

Any other standard component can be simply dragged onto the schematic, and connected.

5.2 Specialised Components

Specialised components are components that were previously special ordered for boards as there were no standard components that suited the use case. Since we generally order spares during PCB orders, often these specialised components are available but rarely used. So if a part is not available as standard, please check the specialised parts.

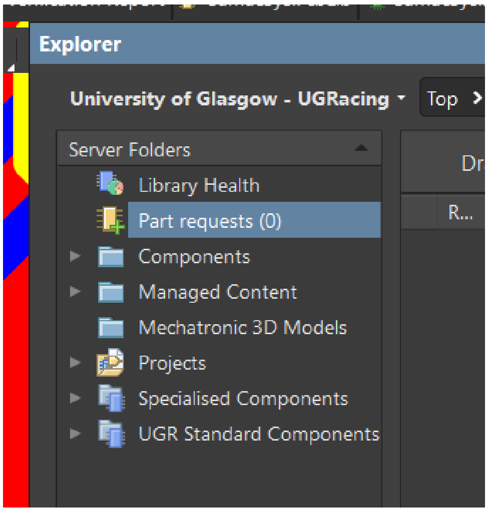
Specialised parts can be found by selecting the category with the suffix "(Specialised)" in the components dropdown.



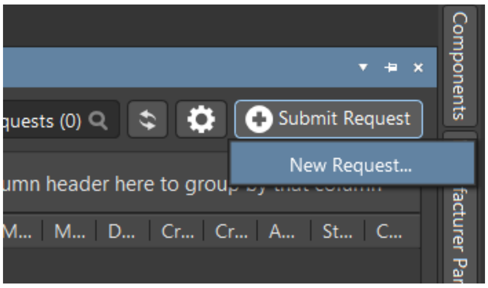
If you cannot find a suitable part in the specialised folder, see section 5.3.

5.3 Part Requests

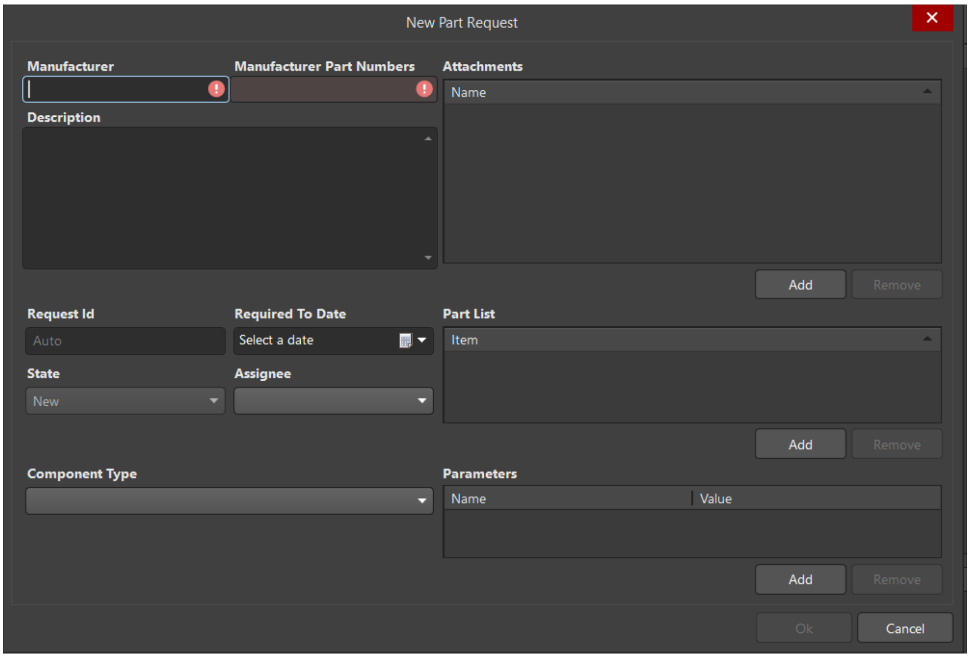
Use part requests if you cannot find a suitable part in the standard or specialised component folders. To submit a part request open the "Explorer" panel on the right hand side (if you can't see it use the "Panels" button).



Open the ”Parts requests” folder, and click ”Submit request”.



Fill out the required information.



> ### Note
> Please let your team head know after you have submitted a part request, it may otherwise go un-noticed.

6 Everything else

Below are a list of YouTube videos that we recommend you use.